UNIT TEST REPORT

Real-Time Chat Application Using Socket Programming

Mini Project Part-3, item 2(b)(i)

Item	Details
Project	Real-Time Chat Application Using Socket Programming
Course	Software Engineering (UE23CS341A), PES University. Mini Project Part-3: Testing
Team	Kabir Raju G (Kabir7code) Keerthana K (KeerthanaK0304) Joshua (jjosh2005)
Repository	github.com/Kabir7code/Chat-application-Socket-Programming-
Test run	11 Nov 2026
Code tested	The complete Sprint 3 code: main on 11 Nov 2026 plus the Sprint 3 pull requests for #53, #54, #59, #60 and #61, which were being merged that day.

1. Purpose and scope

This report gives the results of unit-level testing of the chat application after Sprint 3. It covers two things, both run according to the Test Plan (version 1.1):

- the automated test suite in the tests/ folder (pytest), which is also run by GitHub Actions on every pull request;
- the unit test cases UT_01 to UT_30 in the Test Cases document, each run against the real code and its actual result recorded.

Integration (IT) and system (ST) cases are reported in the System Validation Report; line and branch coverage in the Code Coverage and Branch Coverage Reports.

2. Test environment

Item	Details
Python	3.10.20, 3.12.13 and 3.13.16 (the suite is run on all three)
Operating system	Linux (Ubuntu-based), same as the GitHub Actions runner
Test framework	pytest 9.1, pytest-html 4 (HTML report), JUnit XML output
Static checks	flake8, maximum line length 100 (setup.cfg)
Network	localhost; every test server listens on port 0 (a free port)
Database	SQLite in memory (:memory:) or a temporary file per test

3. Summary

Measure	Result
Automated tests	338 tests in 25 files
Python 3.13	338 passed, 0 failed, 0 skipped (28.2 s)
Python 3.12	338 tests, 0 failures, 0 errors (27.0 s)
Python 3.10	338 tests, 0 failures, 0 errors (27.8 s)
flake8	No warnings on Python 3.10 and 3.12
Unit test cases (UT)	30 run: 28 Pass, 2 Partial, 0 Fail, 0 Not executed
Repeat runs	Every UT case was run 4 times with the same result each time

Result: the automated suite passes completely on every Python version. Of the 30 unit test cases, 28 pass. UT_02 and UT_03 are Partial: the web UI checks usernames before sending, but the terminal client still relies on the server's check (see section 6).

4. Automated tests by file

Test file	What it tests	Tests	Passed	Failed	Time (s)
test_chat_history.py	FR-011 (#56): a new user sees the recent Everyone chat when they log in.	12	12	0	3.55
test_db_setting.py	NFR-004 (#55): where the server keeps its database (--db / CHAT_DB).	7	7	0	0.02
test_delivery_time.py	NFR-001 (#19): a message reaches the other users within 1 second.	3	3	0	0.07
test_group_history.py	FR-010 (#18): someone who joins a group sees its recent messages.	12	12	0	3.59
test_group_history_db.py	FR-010 (#61): group history is read from the database.	8	8	0	1.37
test_groups.py	Group chat management (FR-004). Test case IDs refer to the Test Cases document.	30	30	0	4.27
test_load.py	NFR-002 (#20): the server supports 20 users at the same time.	4	4	0	0.50
test_logging.py	NFR-004 (#14): the server uses the logging module instead of print().	9	9	0	0.09
test_messaging.py	Registration (FR-001), real-time delivery (FR-003) and presence (FR-006).	14	14	0	0.14
test_offline_messages.py	FR-012 (#59): private messages sent while someone is offline.	12	12	0	2.18
test_ordering.py	IT_08 (#31): messages sent at the same moment reach every user in the same order.	3	3	0	0.27
test_persistent_groups.py	FR-013 (#51): groups, their members and their admin survive a restart.	14	14	0	0.21
test_private_history.py	FR-012 (#57): /history <user> shows a private conversation.	12	12	0	0.34
test_protocol.py	Message framing tests (NFR-003).	12	12	0	0.02
test_restart.py	NFR-006 (#60): everything the chat saves survives a server restart.	7	7	0	2.47
test_retention.py	NFR-007 (#54): keep only the most recent messages of each chat.	16	16	0	0.12
test_robustness.py	Disconnects (FR-007), private messages (FR-005), thread safety (NFR-002),	23	23	0	1.59
test_server.py	Server tests for concurrent clients (FR-002).	4	4	0	0.38
test_sprint2_coverage.py	NFR-004 (#12): extra automated tests for the Sprint 2 features that had	23	23	0	0.21
test_storage.py	FR-011 (#50): the persistent message store.	20	20	0	0.25
test_timestamps.py	FR-009 (#13): every chat, group and private message carries the time the	9	9	0	0.16
test_tls.py	TLS encryption (NFR-005 part 2, #22).	11	11	0	1.69
test_users_command.py	FR-006 (#7): /users lists everyone who is online.	10	10	0	0.33
test_web.py	Web gateway (#16), driven over HTTP the way the browser page uses it.	54	54	0	3.30
test_web_history.py	#53: the web UI gets saved history when someone joins.	9	9	0	0.48
Total		338	338	0	27.59

5. Unit test case results (UT_01 to UT_30)

Full steps, test data and expected results are in the Test Cases document.

ID	Module	Description	Actual result	Result
UT_01	User Registration & Connection (FR-001)	Username field accepts a valid alphanumeric username	Terminal client sent the name and the server accepted it; client printed: 'Welcome, alice_23! Online: nobody yet'	**Pass**
UT_02	User Registration & Connection (FR-001)	Empty username is blocked before contacting the Server	Web UI: blocked on the page with 'Use 1-20 letters, digits or underscores.', no request sent. Terminal client: not blocked, the empty name is sent and the server replies '' (#29 is closed, but the terminal client was not changed).	**Partial**
UT_03	User Registration & Connection (FR-001)	Username longer than the maximum allowed length is rejected	Web UI: the name box stops at 20 characters, so the page joined as a 20-character name without any message. Terminal client: not blocked locally; the name is sent and the server rejects it with 'Username must be 1-20 letters,...	**Partial**
UT_04	Multi-User Concurrent Sessions (FR-002)	Session object is created with correct initial state on connect	Session holds username 'testuser1', the socket passed in, and alive=True.	**Pass**
UT_05	Multi-User Concurrent Sessions (FR-002)	Session lookup returns the correct session for a given socket	Lookup returned userA's session only (the registry looks sessions up by username).	**Pass**
UT_06	Multi-User Concurrent Sessions (FR-002)	Removing one session does not alter unrelated sessions	After removing userB the store holds ['userA', 'userC']; userA and userC unchanged.	**Pass**
UT_07	Real-Time Messaging (FR-003)	Message object is correctly constructed from raw input	Function returns {'type': 'chat', 'id': 28, 'time': '...', 'from': 'userA', 'text': 'Hello there'} - sender, text and a generated message ID.	**Pass**
UT_08	Real-Time Messaging (FR-003)	Empty message text is rejected before sending	Pressing Enter on an empty or blank line sends nothing (other users received 0 messages); a blank message sent directly is rejected by the server (EMPTY_MESSAGE).	**Pass**
UT_09	Real-Time Messaging (FR-003)	Message exceeding the maximum length is rejected or truncated per design	Limit is 4000 characters (D4). A 4001-character message is rejected with 'Messages are limited to 4000 characters.' and nothing is delivered; 1500 characters are delivered whole.	**Pass**
UT_10	Group Chat Management (FR-004)	Group object is created with the correct owner and member list	Group 'ProjectTeam' created with members ['userA'] and admin 'userA'.	**Pass**
UT_11	Group Chat Management (FR-004)	Duplicate group name is rejected at creation	Second create rejected with GROUP_EXISTS: 'Group name already exists.'. Only one 'ProjectTeam' exists.	**Pass**
UT_12	Group Chat Management (FR-004)	Membership check correctly identifies whether a user belongs to a group	is_member returns True for userB and False for userC.	**Pass**
UT_13	Private Messaging (FR-005)	Private message object correctly records sender and single recipient	Message built as {'type': 'private', 'from': 'userA', 'to': 'userB', 'text': 'hey', 'id', 'time'}} - one recipient, flagged private.	**Pass**
UT_14	Private Messaging (FR-005)	Private message to a non-existent username is flagged before sending	Server answered 'User not found/offline' (code USER_NOT_FOUND); nothing was delivered or saved.	**Pass**
UT_15	Private Messaging (FR-005)	Private message text is subject to the same validation as other messages	Both rejected by the same check used for group and normal messages: 'Message text is empty.' and 'Messages are limited to 4000 characters.'.	**Pass**
UT_16	Presence & Notifications (FR-006)	Active-user list adds a new entry on connect	'newuser' is now in the active list ['newuser'] and its session is alive.	**Pass**
UT_17	Presence & Notifications (FR-006)	Active-user list removes the correct entry on disconnect	Only userA removed; list is ['userB'].	**Pass**
UT_18	Presence & Notifications (FR-006)	Group member-list update function adds/removes the correct member	Member list is exactly ['userB'] (and userB became admin).	**Pass**
UT_19	Session Termination & Disconnection Handling (FR-007)	Graceful-leave function correctly marks a session as terminated	Session marked not alive and removed from the active list; no exception raised.	**Pass**
UT_20	Session Termination & Disconnection Handling (FR-007)	Unexpected-disconnect detection triggers on a closed socket	Handler saw the closed socket, ended its loop and cleaned up the session (no loop, no crash).	**Pass**
UT_21	Session Termination & Disconnection Handling (FR-007)	Disconnection cleanup releases the correct session resources	Socket handle released (closed) and session removed; userA was taken out of both groups (Squad deleted as empty, Study now ['x']).	**Pass**

ID	Module	Description	Actual result	Result
UT_22	Error Handling (FR-008)	Malformed message payload is caught without an unhandled exception	Invalid bytes, broken JSON and a non-object all raised a handled ProtocolError; no crash.	**Pass**
UT_23	Error Handling (FR-008)	Send failure on one socket returns a handled error, not a crash	send() returned False instead of raising, and the session was marked for cleanup.	**Pass**
UT_24	Error Handling (FR-008)	Error-handling routine logs sufficient detail for diagnosis	A WARNING log entry was written with the time, the user and the error: '2026-11-10 19:57:19,719 WARNING chat.server: Broken frame from alice; disconnecting'. Unexpected errors are logged with a traceback.	**Pass**
UT_25	Persistent Message Store (FR-011)	A saved message is loaded back unchanged	The saved message came back identical: same type, ID, time, sender, text and context.	**Pass**
UT_26	Persistent Message Store (FR-011)	Private history between two users contains only their conversation	alice-bob history is ['hi bob', 'hi alice']: both directions, oldest first, nobody else's messages.	**Pass**
UT_27	Private History & Offline Delivery (FR-012)	The /history command is parsed correctly and misuse is caught locally	'/history bob' becomes {'type': 'private_history', 'with': 'bob'}; '/history' and '/history bob carol' show 'Usage: /history <user>' locally and send nothing. /help lists the command.	**Pass**
UT_28	Persistent Groups (FR-013, FR-010)	Group state is saved and loaded with admin and member order	load_groups() returned {'Squad': ('adam', ['adam', 'mia'])}: the latest admin and members in joining order; the deleted group is gone.	**Pass**
UT_29	Database Settings & Retention (NFR-004, NFR-007)	Database location comes from --db, then CHAT_DB, then the default	Default 'chat.db'; with CHAT_DB set 'env.db'; with --db as well 'flag.db' (the flag wins).	**Pass**
UT_30	Database Settings & Retention (NFR-004, NFR-007)	Invalid --keep values are rejected	--keep -1 and --keep lots are both rejected with a usage error; the default is 500.	**Pass**

6. Defects and observations

Case	Finding	Status
UT_02, UT_03	The terminal client sends an empty or too-long username to the server instead of rejecting it first. The server rejects it with a clear message, so nothing breaks. Backlog item #29 was closed, but the change it describes was never made to chat/client.py (only a .jsx mock-up file was added to the repository root). The web UI blocks an empty name and stops typing at 20 characters.	Open. Reopen #29 (Deviations.md, B4).
UT_07	Fixed since Sprint 1: messages now carry an ID and a timestamp (#30, #13).	Closed
UT_24	Fixed since Sprint 1: errors are logged with time, level, user and traceback (#14).	Closed

7. How to reproduce

```
pip install -r requirements-dev.txt pytest-html
flake8 chat tests
pytest -v --junitxml=junit.xml --html=unit_test_report.html --self-contained-html
```

Attached in Documents/Test_Reports: unit_test_report.html (every test with its result and time), junit.xml (Python 3.13), junit-py3.10.xml and junit-py3.12.xml.

Appendix A. Every automated test

Python 3.13 run. The same tests passed on Python 3.10 and 3.12.

File	Test	Time (s)	Result
test_chat_history	test_history_size_is_20	0.001	Passed

File	Test	Time (s)	Result
test_chat_history	test_first_user_gets_no_history	0.408	Passed
test_chat_history	test_partial_history	0.468	Passed
test_chat_history	test_full_history_keeps_the_last_20_oldest_first	0.472	Passed
test_chat_history	test_history_comes_right_after_registered	0.006	Passed
test_chat_history	test_private_and_group_messages_are_not_in_history	0.511	Passed
test_chat_history	test_live_messages_are_not_marked_history	0.406	Passed
test_chat_history	test_saved_messages_are_not_changed_by_marking	0.025	Passed
test_chat_history	test_history_survives_a_server_restart	0.426	Passed
test_chat_history	test_no_message_is_lost_or_doubled_while_joining	0.821	Passed
test_chat_history	test_terminal_client_shows_earlier	0.001	Passed
test_chat_history	test_terminal_client_live_message_unchanged	0.001	Passed
test_db_setting	test_default_database_file	0.003	Passed
test_db_setting	test_database_from_environment	0.001	Passed
test_db_setting	test_flag_beats_environment	0.001	Passed
test_db_setting	test_memory_database_is_allowed	0.001	Passed
test_db_setting	test_tests_use_a_throwaway_database	0.002	Passed
test_db_setting	test_main_passes_the_database_to_the_server	0.003	Passed
test_db_setting	test_server_creates_the_database_file	0.011	Passed
test_delivery_time	test_broadcast_arrives_within_one_second	0.005	Passed
test_delivery_time	test_private_message_arrives_within_one_second	0.005	Passed
test_delivery_time	test_every_message_in_a_burst_arrives_within_one_second	0.060	Passed
test_group_history	test_history_is_empty_for_a_new_group	0.001	Passed
test_group_history	test_history_keeps_messages_oldest_first	0.000	Passed
test_group_history	test_history_keeps_only_the_most_recent_messages	0.001	Passed
test_group_history	test_history_is_deleted_with_the_group	0.000	Passed
test_group_history	test_history_returns_copies	0.001	Passed
test_group_history	test_new_member_receives_recent_messages	0.492	Passed
test_group_history	test_history_messages_are_marked_and_keep_sender	0.469	Passed
test_group_history	test_added_member_receives_recent_messages	0.449	Passed
test_group_history	test_existing_members_do_not_get_history_again	1.259	Passed
test_group_history	test_new_member_of_an_empty_group_gets_no_history	0.447	Passed
test_group_history	test_only_the_last_messages_are_shown	0.471	Passed
test_group_history	test_client_shows_history_as_earlier	0.001	Passed
test_group_history_db	test_history_comes_from_the_store	0.002	Passed
test_group_history_db	test_record_does_not_keep_a_second_copy_in_memory	0.002	Passed
test_group_history_db	test_store_history_keeps_the_last_10_oldest_first	0.002	Passed
test_group_history_db	test_unknown_group_has_no_history	0.002	Passed
test_group_history_db	test_deleting_a_group_deletes_its_messages	0.002	Passed
test_group_history_db	test_without_a_store_history_stays_in_memory	0.001	Passed
test_group_history_db	test_group_history_survives_a_server_restart	0.510	Passed
test_group_history_db	test_message_sent_while_someone_joins_arrives_once	0.851	Passed
test_groups	test_create_sets_owner_and_members	0.001	Passed
test_groups	test_duplicate_group_name_rejected	0.001	Passed
test_groups	test_membership_check	0.001	Passed
test_groups	test_member_list_add_and_remove	0.001	Passed
test_groups	test_invalid_group_names_rejected	0.001	Passed
test_groups	test_only_admin_can_add	0.001	Passed
test_groups	test_admin_passes_on_and_empty_group_is_deleted	0.001	Passed
test_groups	test_remove_user_takes_them_out_of_every_group	0.001	Passed
test_groups	test_concurrent_creates_of_same_name	0.006	Passed
test_groups	test_create_group	0.004	Passed
test_groups	test_join_group	0.046	Passed
test_groups	test_group_message_reaches_only_members	0.472	Passed
test_groups	test_member_who_leaves_gets_nothing_more	0.447	Passed
test_groups	test_joining_twice_is_harmless	0.025	Passed
test_groups	test_full_group_lifecycle	0.516	Passed
test_groups	test_user_in_two_groups_gets_each_message_under_the_right_group	0.511	Passed
test_groups	test_private_message_not_visible_to_group	0.872	Passed
test_groups	test_private_message_alongside_group_traffic	0.492	Passed
test_groups	test_members_see_updated_list_when_someone_joins	0.050	Passed
test_groups	test_leaving_one_group_keeps_the_other	0.107	Passed
test_groups	test_disconnect_removes_user_from_groups	0.046	Passed
test_groups	test_admin_adds_member	0.067	Passed
test_groups	test_non_admin_cannot_add	0.047	Passed

File	Test	Time (s)	Result
test_groups	test_adding_offline_user	0.025	Passed
test_groups	test_non_member_cannot_send_to_group	0.449	Passed
test_groups	test_unknown_group	0.004	Passed
test_groups	test_group_text_is_validated	0.070	Passed
test_groups	test_unregistered_socket_cannot_use_groups	0.003	Passed
test_groups	test_parse_group_commands	0.001	Passed
test_groups	test_parse_group_commands_need_arguments	0.000	Passed
test_load	test_twenty_users_can_register	0.040	Passed
test_load	test_message_reaches_all_nineteen_others	0.335	Passed
test_load	test_twenty_users_sending_at_once	0.091	Passed
test_load	test_twenty_users_can_be_reached_privately	0.038	Passed
test_logging	test_server_has_no_print_calls	0.001	Passed
test_logging	test_server_uses_its_own_logger	0.000	Passed
test_logging	test_connection_is_logged	0.024	Passed
test_logging	test_registration_is_logged	0.004	Passed
test_logging	test_leaving_is_logged	0.024	Passed
test_logging	test_broken_frame_is_logged_as_warning	0.024	Passed
test_logging	test_unexpected_error_is_logged_with_traceback	0.011	Passed
test_logging	test_log_level_comes_from_environment	0.001	Passed
test_logging	test_unknown_log_level_falls_back_to_info	0.001	Passed
test_messaging	test_unique_username_is_accepted	0.003	Passed
test_messaging	test_duplicate_username_is_rejected	0.003	Passed
test_messaging	test_can_retry_with_another_username	0.004	Passed
test_messaging	test_empty_username_is_rejected	0.003	Passed
test_messaging	test_too_long_username_is_rejected	0.003	Passed
test_messaging	test_username_is_freed_after_leaving	0.054	Passed
test_messaging	test_message_carries_sender_name_and_context	0.005	Passed
test_messaging	test_messages_arrive_in_order	0.052	Passed
test_messaging	test_others_are_told_when_someone_joins	0.004	Passed
test_messaging	test_others_are_told_when_someone_leaves	0.004	Passed
test_messaging	test_registration_reply_lists_online_users	0.004	Passed
test_messaging	test_config_defaults	0.001	Passed
test_messaging	test_config_from_environment	0.001	Passed
test_messaging	test_command_line_overrides_environment	0.001	Passed
test_offline_messages	test_users_are_remembered	0.001	Passed
test_offline_messages	test_waiting_messages_come_out_oldest_first_and_once	0.001	Passed
test_offline_messages	test_waiting_messages_survive_reopening_the_file	0.011	Passed
test_offline_messages	test_sender_is_told_it_will_be_delivered_later	0.026	Passed
test_offline_messages	test_recipient_gets_waiting_messages_after_logging_in	0.428	Passed
test_offline_messages	test_each_waiting_message_is_delivered_once	0.850	Passed
test_offline_messages	test_unknown_name_still_gets_user_not_found	0.004	Passed
test_offline_messages	test_online_messages_are_not_queued	0.048	Passed
test_offline_messages	test_waiting_message_is_kept_in_private_history	0.025	Passed
test_offline_messages	test_nothing_is_lost_while_the_recipient_logs_in	0.333	Passed
test_offline_messages	test_waiting_messages_survive_a_server_restart	0.454	Passed
test_offline_messages	test_client_shows_queued_and_offline_messages	0.001	Passed
test_ordering	test_all_receivers_see_the_same_order	0.115	Passed
test_ordering	test_each_senders_messages_stay_in_sending_order	0.050	Passed
test_ordering	test_group_members_see_the_same_order	0.108	Passed
test_persistent_groups	test_group_state_round_trip	0.001	Passed
test_persistent_groups	test_saving_again_replaces_the_old_state	0.001	Passed
test_persistent_groups	test_delete_group	0.001	Passed
test_persistent_groups	test_members_keep_their_joining_order	0.001	Passed
test_persistent_groups	test_created_group_is_restored	0.011	Passed
test_persistent_groups	test_joined_and_added_members_are_restored	0.010	Passed
test_persistent_groups	test_leaving_is_saved	0.011	Passed
test_persistent_groups	test_new_admin_is_saved	0.010	Passed
test_persistent_groups	test_empty_group_is_deleted_from_the_database	0.012	Passed
test_persistent_groups	test_rules_still_apply_after_a_restart	0.012	Passed
test_persistent_groups	test_without_a_database_nothing_is_saved	0.000	Passed
test_persistent_groups	test_group_survives_a_server_restart	0.037	Passed
test_persistent_groups	test_admin_can_still_add_after_a_restart	0.022	Passed
test_persistent_groups	test_user_who_quits_still_leaves_their_groups	0.081	Passed
test_private_history	test_history_shows_both_directions_oldest_first	0.064	Passed

File	Test	Time (s)	Result
test_private_history	test_nobody_else_can_read_it	0.050	Passed
test_private_history	test_only_the_last_20_are_shown	0.058	Passed
test_private_history	test_other_conversations_are_left_out	0.094	Passed
test_private_history	test_empty_history	0.004	Passed
test_private_history	test_bad_name_is_an_error	0.003	Passed
test_private_history	test_must_be_registered	0.003	Passed
test_private_history	test_history_works_after_a_restart	0.063	Passed
test_private_history	test_parse_history_command	0.000	Passed
test_private_history	test_history_is_in_help	0.000	Passed
test_private_history	test_client_shows_the_conversation	0.000	Passed
test_private_history	test_client_shows_empty_history	0.000	Passed
test_protocol	test_round_trip	0.001	Passed
test_protocol	test_long_message_arrives_whole	0.001	Passed
test_protocol	test_rapid_messages_stay_separate	0.001	Passed
test_protocol	test_split_frame_is_reassembled	0.001	Passed
test_protocol	test_unicode	0.001	Passed
test_protocol	test_closed_connection_returns_none	0.001	Passed
test_protocol	test_invalid_json_raises	0.001	Passed
test_protocol	test_missing_type_raises	0.000	Passed
test_protocol	test_oversized_frame_is_rejected	0.001	Passed
test_protocol	test_make_chat_message	0.001	Passed
test_protocol	test_message_ids_are_unique_and_increasing	0.001	Passed
test_protocol	test_message_ids_are_unique_across_threads	0.006	Passed
test_restart	test_everyone_history_survives	0.465	Passed
test_restart	test_private_history_survives	0.056	Passed
test_restart	test_offline_messages_survive	0.442	Passed
test_restart	test_groups_survive	0.072	Passed
test_restart	test_group_history_survives	0.490	Passed
test_restart	test_everything_survives_two_restarts	0.917	Passed
test_restart	test_memory_database_keeps_nothing	0.025	Passed
test_retention	test_everyone_chat_keeps_the_newest	0.002	Passed
test_retention	test_nothing_is_removed_below_the_limit	0.001	Passed
test_retention	test_each_private_pair_has_its_own_limit	0.001	Passed
test_retention	test_each_group_has_its_own_limit	0.001	Passed
test_retention	test_chats_do_not_trim_each_other	0.001	Passed
test_retention	test_zero_keeps_everything	0.002	Passed
test_retention	test_limit_applies_to_an_existing_file	0.023	Passed
test_retention	test_default_limit_is_500	0.001	Passed
test_retention	test_limit_from_environment	0.001	Passed
test_retention	test_flag_beats_environment	0.001	Passed
test_retention	test_bad_limit_is_rejected[-1]	0.001	Passed
test_retention	test_bad_limit_is_rejected[lots]	0.001	Passed
test_retention	test_bad_limit_is_rejected[2.5]	0.001	Passed
test_retention	test_bad_flag_stops_the_server	0.001	Passed
test_retention	test_server_passes_the_limit_to_the_store	0.001	Passed
test_retention	test_server_trims_while_running	0.078	Passed
test_robustness	test_quit_removes_user_and_notifies_others	0.009	Passed
test_robustness	test_crashed_client_is_cleaned_up	0.006	Passed
test_robustness	test_server_keeps_working_after_a_crash	0.009	Passed
test_robustness	test_user_can_reconnect_after_crash	0.025	Passed
test_robustness	test_keepalive_is_enabled_on_client_sockets	0.007	Passed
test_robustness	test_private_message_reaches_only_the_recipient	0.309	Passed
test_robustness	test_sender_sees_confirmation	0.046	Passed
test_robustness	test_private_message_to_unknown_user	0.004	Passed
test_robustness	test_private_message_to_user_who_left	0.027	Passed
test_robustness	test_only_one_of_many_simultaneous_registrations_wins	0.024	Passed
test_robustness	test_concurrent_senders_never_corrupt_messages	0.113	Passed
test_robustness	test_invalid_json_gets_error_and_session_continues	0.006	Passed
test_robustness	test_oversized_frame_disconnects_only_that_client	0.026	Passed
test_robustness	test_unknown_message_type	0.004	Passed
test_robustness	test_empty_message_is_rejected	0.306	Passed
test_robustness	test_wrong_field_types_do_not_crash_session	0.049	Passed
test_robustness	test_unregistered_socket_cannot_chat	0.305	Passed
test_robustness	test_unregistered_socket_cannot_send_private	0.306	Passed

File	Test	Time (s)	Result
test_robustness	test_cannot_register_twice	0.006	Passed
test_robustness	test_parse_plain_text	0.000	Passed
test_robustness	test_parse_private_message	0.001	Passed
test_robustness	test_parse_private_message_without_text	0.000	Passed
test_robustness	test_parse_quit_and_unknown	0.000	Passed
test_server	test_many_clients_connect_at_once	0.056	Passed
test_server	test_message_reaches_other_clients	0.006	Passed
test_server	test_sender_does_not_get_own_message	0.307	Passed
test_server	test_new_client_does_not_disturb_existing_session	0.006	Passed
test_sprint2_coverage	test_group_messages_get_an_id	0.049	Passed
test_sprint2_coverage	test_ids_increase_across_chat_private_and_group	0.091	Passed
test_sprint2_coverage	test_private_confirmation_has_same_id_as_delivered_message	0.046	Passed
test_sprint2_coverage	test_new_message_id_is_unique_under_threads	0.013	Passed
test_sprint2_coverage	test_format_message[message0-alice: hi]	0.001	Passed
test_sprint2_coverage	test_format_message[message1-[Squad] alice: hi]	0.001	Passed
test_sprint2_coverage	test_format_message[message2-[private] bob: psst]	0.001	Passed
test_sprint2_coverage	test_format_message[message3-[private to bob] psst]	0.001	Passed
test_sprint2_coverage	test_format_message[message4-* bob joined the chat]	0.001	Passed
test_sprint2_coverage	test_format_message[message5-! Username already taken.]	0.001	Passed
test_sprint2_coverage	test_format_message[message6-* [Squad] bob joined Squad (members: alice, bob)]	0.001	Passed
test_sprint2_coverage	test_format_message[message7-* [Squad] You left Squad (members: none)]	0.001	Passed
test_sprint2_coverage	test_help_lists_every_command	0.000	Passed
test_sprint2_coverage	test_group_message_keeps_spaces_in_text	0.000	Passed
test_sprint2_coverage	test_add_needs_group_and_user	0.001	Passed
test_sprint2_coverage	test_slash_alone_is_unknown	0.001	Passed
test_sprint2_coverage	test_tls_is_on_by_default	0.001	Passed
test_sprint2_coverage	test_no_tls_from_environment	0.001	Passed
test_sprint2_coverage	test_cert_paths_can_be_overridden	0.001	Passed
test_sprint2_coverage	test_client_without_cafile_gives_clear_message	0.002	Passed
test_sprint2_coverage	test_registry_register_and_disconnect	0.000	Passed
test_sprint2_coverage	test_registry_disconnect_of_unregistered_session	0.000	Passed
test_sprint2_coverage	test_registry_usernames_are_sorted	0.000	Passed
test_storage	test_new_store_is_empty	0.001	Passed
test_storage	test_chat_message_round_trip	0.001	Passed
test_storage	test_private_message_round_trip	0.001	Passed
test_storage	test_group_message_round_trip	0.001	Passed
test_storage	test_messages_come_back_oldest_first	0.002	Passed
test_storage	test_limit_keeps_the_most_recent	0.008	Passed
test_storage	test_kinds_are_kept_apart	0.002	Passed
test_storage	test_private_history_has_both_directions_and_nobody_else	0.001	Passed
test_storage	test_groups_are_kept_apart	0.001	Passed
test_storage	test_tables_are_created_in_a_new_file	0.011	Passed
test_storage	test_messages_survive_reopening_the_file	0.011	Passed
test_storage	test_safe_to_use_from_many_threads	0.062	Passed
test_storage	test_a_storage_error_is_logged_not_raised	0.002	Passed
test_storage	test_server_saves_broadcast_messages	0.005	Passed
test_storage	test_server_saves_delivered_private_messages	0.049	Passed
test_storage	test_private_message_to_unknown_user_is_not_saved	0.004	Passed
test_storage	test_server_saves_group_messages	0.050	Passed
test_storage	test_rejected_messages_are_not_saved	0.003	Passed
test_storage	test_saved_order_matches_delivery_order	0.022	Passed
test_storage	test_server_with_a_database_file	0.013	Passed
test_timestamps	test_timestamp_is_utc_iso_8601	0.001	Passed
test_timestamps	test_chat_message_has_a_time	0.001	Passed
test_timestamps	test_client_shows_local_hours_and_minutes	0.001	Passed
test_timestamps	test_group_and_private_lines_get_the_time_first	0.000	Passed
test_timestamps	test_missing_or_bad_time_shows_no_prefix	0.000	Passed
test_timestamps	test_broadcast_arrives_with_a_time	0.004	Passed
test_timestamps	test_private_message_and_confirmation_share_a_time	0.048	Passed
test_timestamps	test_group_message_arrives_with_a_time	0.047	Passed
test_timestamps	test_times_do_not_go_backwards	0.055	Passed
test_tls	test_chat_works_over_tls	0.366	Passed
test_tls	test_captured_traffic_has_no_plaintext	0.343	Passed
test_tls	test_control_without_tls_the_tap_does_see_plaintext	0.250	Passed

File	Test	Time (s)	Result
test_tls	test_client_rejects_server_it_does_not_trust	0.270	Passed
test_tls	test_plain_client_cannot_talk_to_tls_server	0.050	Passed
test_tls	test_stalled_handshake_does_not_block_others	0.050	Passed
test_tls	test_groups_work_over_tls	0.140	Passed
test_tls	test_cert_covers_extra_host	0.001	Passed
test_tls	test_missing_cert_gives_clear_message	0.003	Passed
test_tls	test_no_tls_flag	0.002	Passed
test_tls	test_real_terminal_client_over_tls	0.216	Passed
test_users_command	test_parse_users_command	0.000	Passed
test_users_command	test_users_is_in_help	0.000	Passed
test_users_command	test_format_users_reply	0.000	Passed
test_users_command	test_lists_everyone_online_sorted	0.006	Passed
test_users_command	test_only_user_sees_themselves	0.003	Passed
test_users_command	test_user_who_left_is_no_longer_listed	0.004	Passed
test_users_command	test_new_user_appears_in_list	0.005	Passed
test_users_command	test_connected_but_unregistered_users_are_not_listed	0.004	Passed
test_users_command	test_unregistered_socket_cannot_list_users	0.003	Passed
test_users_command	test_reply_goes_only_to_the_asker	0.307	Passed
test_web	test_two_users_connect	0.094	Passed
test_web	test_duplicate_username_is_refused_and_can_retry	0.065	Passed
test_web	test_invalid_username_is_refused[]	0.055	Passed
test_web	test_invalid_username_is_refused[has space]	0.056	Passed
test_web	test_invalid_username_is_refused[xxxxxxxxxxxxxxxxxxxxx]	0.055	Passed
test_web	test_invalid_username_is_refused[bad!name]	0.054	Passed
test_web	test_username_must_be_text	0.055	Passed
test_web	test_chat_server_unreachable	0.055	Passed
test_web	test_broadcast_reaches_the_other_stream	0.066	Passed
test_web	test_sender_sees_their_own_message_once	0.067	Passed
test_web	test_refused_message_is_not_echoed[ ]	0.065	Passed

File	Test	Time (s)	Result
test_web	test_refused_message_is_not_echoed[xxxx

File	Test	Time (s)	Result
test_web	test_unknown_session_is_404[/api/disconnect-payload2]	0.056	Passed
test_web	test_unknown_session_is_404[/api/send-payload3]	0.056	Passed
test_web	test_events_with_unknown_session_is_404	0.055	Passed
test_web	test_send_needs_a_proper_message[None]	0.061	Passed
test_web	test_send_needs_a_proper_message[text]	0.066	Passed
test_web	test_send_needs_a_proper_message[message2]	0.061	Passed
test_web	test_send_needs_a_proper_message[message3]	0.060	Passed
test_web	test_oversized_message_is_refused	0.060	Passed
test_web	test_post_from_another_site_is_refused	0.054	Passed
test_web	test_bad_json_is_refused	0.054	Passed
test_web	test_huge_body_is_refused	0.054	Passed
test_web	test_page_files_are_served[/-text/html-join-form]	0.054	Passed
test_web	test_page_files_are_served[/index.html-text/html-join-form]	0.055	Passed
test_web	test_page_files_are_served[/app.js-text/javascript-EventSource]	0.058	Passed
test_web	test_page_files_are_served[/style.css-text/css-.messages]	0.055	Passed
test_web	test_favicon_request_is_answered_quietly	0.055	Passed
test_web	test_nothing_else_can_be_fetched[/../setup.cfg]	0.056	Passed
test_web	test_nothing_else_can_be_fetched[/..%2fsetup.cfg]	0.055	Passed
test_web	test_nothing_else_can_be_fetched[/%2e%2e/setup.cfg]	0.055	Passed
test_web	test_nothing_else_can_be_fetched[/static/../../setup.cfg]	0.054	Passed
test_web	test_nothing_else_can_be_fetched[//etc/passwd]	0.057	Passed
test_web	test_nothing_else_can_be_fetched[/web.py]	0.055	Passed
test_web	test_nothing_else_can_be_fetched[/chat/web.py]	0.054	Passed
test_web	test_gateway_binds_to_localhost_by_default	0.002	Passed
test_web	test_gateway_flags_and_chat_server_flags	0.001	Passed
test_web_history	test_everyone_history_reaches_the_page	0.066	Passed
test_web_history	test_page_can_load_a_private_chat	0.068	Passed
test_web_history	test_history_command_works_from_the_page	0.066	Passed
test_web_history	test_other_peoples_private_chats_stay_private	0.072	Passed
test_web_history	test_group_history_reaches_the_page	0.074	Passed
test_web_history	test_page_gets_its_groups_back_after_a_server_restart	0.130	Passed
test_web_history	test_page_asks_for_private_history_when_a_chat_opens	0.001	Passed
test_web_history	test_page_handles_private_history_replies	0.000	Passed
test_web_history	test_page_does_not_count_old_messages_as_unread	0.001	Passed